Niccolò Covoni

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Research Profile

Philosophy of Information, Philosophy of Physics, Metaphysics of Science, and Philosophy of Science.

Academic Position

2027 - 2030	**Fixed-Term Assistant Professor**, Sophia University Institute	*Loppiano, Italy*

Education

PhD candidate in Philosophy and in Research Methods in Science and Technology, Università della Svizzera italiana (USI) and University of Urbino Carlo Bo, 2023 - present. Dissertation: *The Reality of Entropy: A Foundational Study of Entropy and Information*. Supervisors: Professor Claudio Calosi and Professor Vincenzo Fano. **Expected thesis defense: February 2027**.

MS in Logic, Philosophy and History of Science, University of Florence, 2023, 110/110 cum laude. Dissertation: *Methods of Abstraction in Information Theory: The Case of Quantum-Like Machine Learning*.

BS in Philosophy, University of Pisa, 2020. Dissertation on the foundations of Birkhoff-von Neumann quantum logic.

Research Visits

07/26 - 08/26	**FTG Visiting Fellow**, Foundations of Thermodynamics Group, Nanyang Technological University; hosted by Professor Eugene Y. S. Chua	*Singapore*
02/26 - 04/26	**Visiting Student**, University of Lodz; supervised by Professor Hanna Podsędkowska	*Lodz, Poland*
01/25 - 03/25	**Visiting Student**, Rotman Institute of Philosophy, Western University; QISS project, supervised by Professor Francesca Vidotto	*Canada*
2024	**Visiting Student**, University of Cagliari; supervised by Professors Roberto Giuntini and Giuseppe Sergioli	*Cagliari, Italy*

Prizes & Grants

2025	**QISS Consortium Scholarship**, Western University	*Canada*
2023 - present	**PhD Scholarship**, Department of Pure and Applied Sciences, University of Urbino Carlo Bo	*Italy*
2020 - 2022	**Excellence Scholarship**, Percorso di Eccellenza Trasversale, University of Florence	*Italy*

Selected Publications

Books

Carlo Rovelli, Niccolò Covoni, Andrea Di Biagio. *Tractatus Quanticus*. Wiley. Forthcoming April 2027.

Niccolò Covoni, Andrea Di Biagio, Carlo Rovelli. *Tractatus Quanticus*. Adelphi. Italian edition under contract.

Articles, Chapters, and Proceedings

Niccolò Covoni, Marco Sanchioni. Forthcoming. Entanglement, Relative Facts, and the Meaning of Information in Relational Quantum Mechanics. *European Journal for Philosophy of Science*.

Niccolò Covoni, Marco Sanchioni. 2026. Relationality without Agents: Quantum Correlations and the Limits of QBism. In Anguel S. Stefanov and Vera Lyubenova (eds.), *Niels Bohr and Contemporary Philosophy of Science*, Minkowski Institute Press, pp. 61-86.

Vincenzo Fano, Niccolò Covoni. 2026. Phenomenological Elements in Carnap's *Aufbau*: The Epistemological Role of Sensible Intuition. *Sophia*, XVIII (2026-1), 29-51.

Niccolò Covoni, Marco Sanchioni. 2026. Relational Information Towards a New Kind of Information in Quantum Mechanics. *Foundations of Physics*, 56(1): 14.

Niccolò Covoni, Vincenzo Fano, Marco Sanchioni. 2026. Relational Quantum Mechanics and (Soft) Perspectivism. *Foundations of Physics*, 56(1): 6.

Niccolò Covoni, Flavia Marcacci, Gino Tarozzi. 2025. From Neo-Idealistic Historicism to Geymonat's Materialistic and Agazzi's Objectivistic Realism: The Roots of the Italian Philosophy of Science. *Bioethics Update*, 11(2), 129-140.

Niccolò Covoni. 2025. From Shannon to von Neumann: A Partial Understanding of Information. *Atti del XLIV Congresso Nazionale SISFA - Firenze 2024*, 89-94.

Niccolò Covoni, Alberto Corti, Vincenzo Fano. 2024. A Model-Based Form of Naturalised Metaphysics. *Revista Portuguesa de Filosofia*, 80(4), 783-818.

Niccolò Covoni, Gino Tarozzi. 2024. Empiricism and Realism Reconciled in Agazzi's Conception of Scientific Objectivity. *Distinctio: Journal of Intersubjective Studies*, 3(1), 47-55.

Niccolò Covoni, Giovanni Galli. 2025. A (Quasi-)Mathematical Theory of Public Language. *Proceedings of the 46th International Wittgenstein Symposium*.

Niccolò Covoni, Giovanni Macchia, Davide Pietrini, Gino Tarozzi. 2025. Non-Standard Realistic Models of Quantum Phenomena and New Forms of Complementarity. *Comptes Rendus de l'Académie Internationale de Philosophie des Sciences*, 3, 39-54.

Niccolò Covoni, Silvano Zipoli Caiani. 2024. Levels of Abstraction as Family Resemblances from the Classical to the Quantum Mechanical Representation of Reality. *Proceedings of the 45th International Wittgenstein Symposium*.

Selected Work in Review

Niccolò Covoni, Hanna Podsędkowska, Rafał Wieczorek. 2026. Segal Entropy and Relational Information.

Niccolò Covoni, Marco Sanchioni. 2026. From "It from Bit" to Relational Quantum Mechanics: Wheeler, Rovelli, and the Relational Turn in Quantum Foundations.

Selected Presentations

Invited Talks

27–30/05/2027. **Invited speaker, Foundations of Quantum Theory session**. New Directions in the Foundations of Physics, Hotel Bohinj, Slovenia.

23/04/2026. **Relational Information Towards a New Kind of Information in Quantum Mechanics**. Universidad Complutense de Madrid, Spain.

19/04/2026. **Entanglement, fatti relativi e il significato di "informazione" nell'interpretazione relazionale della meccanica quantistica**. University of Urbino, Cesena, Italy.

11/03/2026. **Relational Information Towards a New Kind of Information in Quantum Mechanics**. Jagiellonian University, Krakow, Poland.

05/03/2026. **Relational Information Towards a New Kind of Information in Quantum Mechanics**. University of Lodz, Poland.

17/12/2025. **Once Again on von Neumann Entropy: Is It Always the Case That von Neumann Entropy Corresponds to Thermodynamic Entropy?**. Pontificia Universidad Católica de Valparaíso, Chile.

20/11/2025. **Relational Information: Towards a New Type of Information in Quantum Mechanics**. Institute for Quantum Optics and Quantum Information, Vienna, Austria.

Contributed Presentations

23/09/2026. **Entanglement, Relative Facts, and the Meaning of Information in Relational Quantum Mechanics**. The Warsaw Dialogues in the Philosophy of Physics 2026, Warsaw University of Technology, Warsaw, Poland.

22/07/2026. **Relational Information Towards a New Kind of Information in Quantum Mechanics**. Asian Philosophy of Science Association Conference, NTU Singapore.

29/06/2026. **Entanglement, Relative Facts, and the Meaning of Information in Relational Quantum Mechanics**. SILFS Triennial International Conference, IUSS Pavia, Italy.

09/05/2026. **Segal Entropy and the Notion of Information**. Perspectives on the Foundations of Information Theory and Statistical Mechanics, USI, Lugano.

10/12/2025. **From "It from Bit" to Relational Quantum Mechanics: Wheeler and Rovelli in Historical and Conceptual Perspective**. Towards a Quantum Culture, University of Milan, Italy.

16/10/2025. **Relational Quantum Mechanics and (Soft) Perspectivism**. Philosophy of Science Conference for the 140th Anniversary of Niels Bohr's Birth, Sofia, Bulgaria.

Teaching

2026/2027	**Lugano Philosophy Colloquia**, Course for master's students	*USI, Lugano*
2026/2027	**Philosophy of Language**, Course for bachelor's students	*Sophia*
2026/2027	**Logic, Game Theory and Happiness**, Course for master's students; co-taught with Marco Sanchioni	*Sophia*
2026	**Introduction to the Philosophy of Thermodynamics**, Introductory course for philosophy students	*USI, Lugano*
2025/2026	**Philosophy of Language**, Teaching Assistant to Professor Marco Sanchioni	*Sophia*
2024	**Epistemology**, Teaching Assistant to Professor Gino Tarozzi	*Urbino*
2024	**Introduction to Edutainment**, Lecture for master's students in Communication Science	*Florence*
2023	**Introduction to Description Logic**, Lecture for master's students in Philosophy	*Urbino*

Academic Service

Coordinator, CI^2 - Complexity & Information Initiative, ICTS, Sophia University Institute; with Marco Sanchioni as co-coordinator, 2026 - present.

Editor, Scientific Committee, *Sophia: Research on Fundamentals and Correlation of Knowledge*.

Referee for *Foundations of Science*, *Foundations of Physics*, *Synthese*, *Isonomia*, *The Reasoner*, *Humana.mente*, *The European Physical Journal H*, *The British Journal for the Philosophy of Science*, *Humanities and Social Sciences Communications*, *Studies in History and Philosophy of Science*, and *Philosophical Inquiries*.

Organiser, Perspectives on the Foundations of Information Theory and Statistical Mechanics, USI, Lugano, 2026.

Organiser, International Summer School in Philosophy of Physics, University of Urbino, since 2024.